

TARGET ARTICLE WITH COMMENTARY AND RESPONSE

Listening to language at birth: evidence for a bias for speech in neonates

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For a commentary on this article see Rosen and Iverson (2007).

Abstract

The nature and origin of the human capacity for acquiring language is not yet fully understood. Here we uncover early roots of this capacity by demonstrating that humans are born with a preference for listening to speech. Human neonates adjusted their high amplitude sucking to preferentially listen to speech, compared with complex non-speech analogues that controlled for critical spectral and temporal parameters of speech. These results support the hypothesis that human infants begin language acquisition with a bias for listening to speech. The implications of these results for language and communication development are discussed.

Introduction

Learning how to communicate depends crucially on the ability to select meaningful signals from the environment. For human infants, this requires selectively attending to those auditory (or visual) units that carry communicative content, a problem made complex by the richness of the infant's world. Many animals filter their rich acoustic world through a general predisposition for the vocalizations of conspecifics (members of the same species), a selectivity which, in some cases, is evident even at birth (e.g. Marler, 1990). Do humans show a similar early bias for listening to speech? A bias for listening to speech would provide a potential sieve through which newborns could glean the acoustic signals important for communication.

At birth, infants already have a remarkable facility for discriminating and categorizing many aspects of human language. For example, newborns are sensitive to word boundaries (Christophe, Dupoux, Bertoncini & Mehler, 1994), distinguish between rhythmically dissimilar languages (Mehler, Jusczyk, Lambertz, Halsted, Bertoncini & Amiel-Tison, 1988; Nazzi, Bertoncini & Mehler, 1998; Ramus, Hauser, Miller, Morris & Mehler, 2000), distinguish between stress patterns of multisyllabic words (Sansavini, Bertoncini & Giovanelli, 1997), categorically discriminate lexical versus grammatical words (Shi, Werker & Morgan, 1999), and differentiate between good and poor syllable

forms (Bertoncini & Mehler, 1981). Moreover, infants respond differentially to speech and non-speech. Neonates are able to discriminate languages from different rhythmical classes when the speech is played forwards, but not when it is played backwards (Ramus *et al.*, 2000), suggesting that this ability is based on particular properties of speech, and not applicable to just any patterned complex sound. Although at least some of these perceptual abilities may not be unique to humans; for example, both rats (Toro, Trobalon & Sebastián-Gallés, 2005) and tamarin monkeys (Ramus *et al.*, 2000) can discriminate languages from rhythmical classes in forward but not backwards speech, only humans learn language, suggesting that some aspect(s) of the acquisition process must be unique to humans (for candidates see Pinker & Jackendoff, 2005; Werker & Vouloumanos, 2000). Moreover, speech and non-speech are represented in different areas of the brain in humans: Neuroimaging studies demonstrate that listening to forward speech activates different areas of the infant brain than does backwards speech, in both neonates (Peña, Maki, Kovacic, Dehaene-Lambertz, Koizumi, Bouquet & Mehler, 2003) and 3-month-old infants (Dehaene-Lambertz, Dehaene & Hertz-Pannier, 2002), though the areas of differential activation differ in these two studies, suggesting that the neonatal brain already discriminates between speech and non-speech sounds.

Despite evidence for differential processing for speech and non-speech, a behavioural preference for the speech

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signal itself has yet to be demonstrated in the neonatal period. As Doupe and Kuhl (1999) note: 'In humans, there is no convincing experimental evidence that infants have an innate description of speech'. In an often-cited methodological study, neonates favoured a stimulus that included a speech component (folk music) over a non-speech condition deliberately made unappealing (broadband white noise – the unmodulated sound of radio static) (Butterfield & Siperstein, 1970). On the basis of this study, it was widely reported that neonates prefer speech. However, the differences in spectral and temporal parameters between speech and white noise (modulated vs. invariant signals) (Eisenberg, 1976), and the choice of a 'speech' condition that includes both vocal and musical components, leave this question unanswered.

To investigate whether neonates demonstrate a bias for speech, we presented infants with isolated syllables of human speech contrasted with non-speech stimuli crafted to control for infants' sensitivity to critical spectral and temporal parameters of speech. These stimuli had been used in previous studies investigating listening preferences in infancy, in which we demonstrated that infants as young as 2 months old prefer listening to speech (Vouloumanos & Werker, 2004). The speech signal is composed of concentrations of energy at multiple frequencies that change over time (Figure 1C). Non-speech counterparts were modelled on sine-wave analogues of speech (Remez, Rubin, Pisoni & Carrell, 1981), and consisted of time-varying sinusoidal waves that track the resonant centre frequencies (formants) of natural speech to reproduce the changes in these frequency peaks across time (Figure 1B). In reproducing the main spectral and temporal changes in natural speech, these complex non-speech analogues contrast sharply with single-frequency tones (Figure 1A) and white noise (Figure 1D), two types of stimuli commonly used as non-speech conditions (e.g. Butterfield & Siperstein, 1970; Eisenberg, 1976). In the present study, we investigate whether human neonates show a bias for listening to speech by comparing neonates' contingent sucking responses in eliciting speech and complex non-speech sounds.

Method

Participants

Twenty-two neonates (1–4 days old, $M = 45.1$ hr) were recruited from a local hospital and tested in a high amplitude sucking (HAS) procedure (Cooper & Aslin, 1990; Eimas, Siqueland, Jusczyk & Vigorito, 1971). An additional 24 infants were not included for the following reasons: falling asleep (1), failing to meet the sucking criterion (8; see below), equipment failure (1), experimenter interference (5), crying or fussing (2), rejection of pacifier

(3), sucking weakly (2) and hospital fire alarm ringing during the experiment (2).

Stimuli

Speech stimuli

Speech stimuli consisted of four tokens of a monosyllabic nonsense word ('lif') spoken by a female native English speaker. Tokens varied in intonational contour (average minimum and maximum pitches = 203 Hz, 325 Hz), and in duration (average = 665 ms). The limited variation in phonetic information minimized differences between the speech and non-speech stimulus sets.

Complex non-speech analogues

Non-speech stimuli consisted of time varying sinusoidal waves tracking the main regions of significant energy, namely the fundamental frequency and the first three formants of speech (stimulus creation has been reported in Vouloumanos, Kiehl, Werker & Liddle, 2001; Vouloumanos & Werker, 2004). Non-speech analogues retained the duration, pitch contour, amplitude envelope, relative formant amplitude, and relative intensity of their speech counterparts. The two stimulus types differed in voice quality (non-speech analogues have none), in naturalness or biological quality (non-speech analogues are artifacts), and in the characteristics of the source (speech has one source, the vocal tract, while non-speech analogues have four, one per sinusoidal tone). However, and crucially for the question asked in the current study, the non-speech analogues track changes across time for the peak frequencies of their speech counterparts, and in so doing, follow very closely the spectral and timing changes of natural speech. The signals were further equated for infant ears by retaining the fundamental frequency that carries information about pitch contour of the speech counterparts, because pitch contour contributes importantly to infants' preference for infant-directed speech (Fernald & Kuhl, 1987), and discrimination of their native language (Nazzi, Floccia & Bertoncini, 1998).¹

Design and procedure

Approximately 2 hours after feeding, neonates were presented with a sterilized pacifier coupled to a pressure

¹ The addition of the fundamental frequency is detrimental to the perception of traditional sine-wave analogues (Remez & Rubin, 1993). However, omitting this pitch information from the non-speech analogues would render the comparison trivial for infants, since this dimension alone would predispose the infants towards speech.

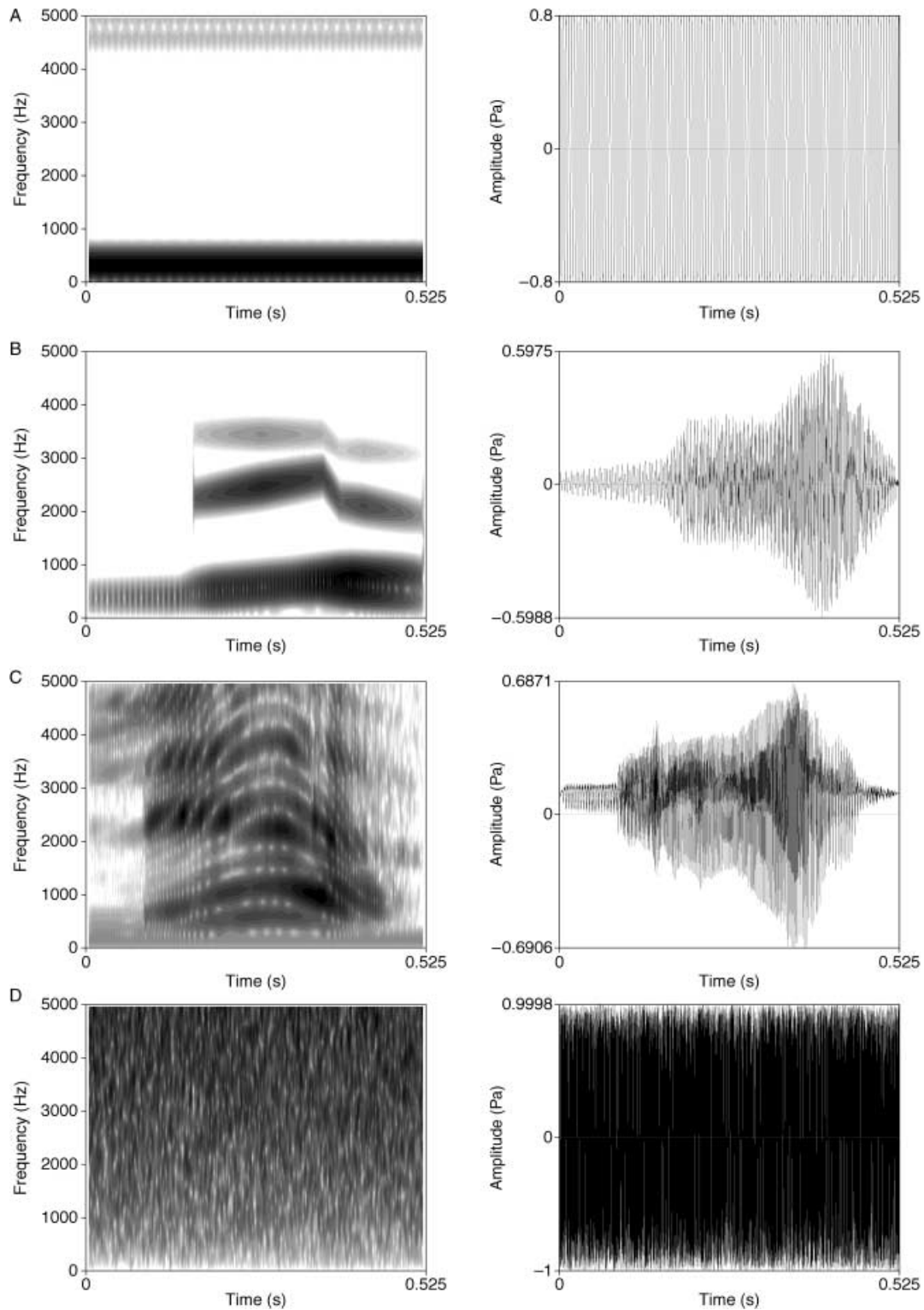


Figure 1 Comparison of acoustic properties of speech and non-speech stimuli. Wide-band spectrograms (left) depict the change in frequency across time, and waveform diagrams (right) illustrate the amplitude changes across time in pressure units; (A) single frequency tones, (B) sample token of complex non-speech used in this study, (C) sample speech token used in this study, and (D) white noise. The changes in frequency across time of the speech signal can also be observed in the complex non-speech sounds whereas this time-varying property is absent from the other sounds.

transducer. Following one silent baseline minute, during which each individual infant's sucking amplitude range was established, neonates were presented with a sound stimulus every time they delivered suction in the upper 80% of their sucking amplitude range. The presentation of speech and non-speech stimuli alternated every minute, and the 8 minutes post-baseline were submitted to analysis.

To ensure that infants had enough exposure to hear the different sounds and learn the contingency, it was necessary to implement criteria for an acceptable minimum number of HA sucks. First, infants were excluded from the analysis if they had any experimental minutes in which they delivered 0 HA sucks. This ensured that newborns whose data were analyzed would hear at least one sound in each experimental minute. Second, in order to ensure that enough sounds were heard at the beginning of the study for the infants to demonstrate a potential preference, we excluded infants who delivered fewer than 10 HA sucks in each of the first four experimental minutes (Flocchia, Nazzi & Bertoncini, 2000).

Results

Based on previous studies with neonates using this HAS procedure in which differences emerged in the latter part of the experiment (Flocchia *et al.*, 2000; Sansavini *et al.*, 1997), the experimental phase was examined as two blocks. A 2 (sound type: speech vs. non-speech) $\times$ 2 (stimulus block: first 4-min block vs. second 4-min block) $\times$ 2 (order: speech first vs. non-speech first) mixed analysis of variance (ANOVA) indicated no main effect of sound type and no main effect of order. A main effect of block ($F(1, 20) = 4.70$, $p = .042$) revealed a higher number of sucks in the first block ($M = 74.4$, $SE = 4.52$) than the second block ($M = 68.2$, $SE = 3.51$) overall. There was no main effect of sound type; however, there was a significant interaction between sound type and experimental block ($F(1, 20) = 6.68$, $p = .018$); planned comparisons on sucking rates in each of the two blocks showed that neonates sucked significantly more to listen to speech ($M = 73.0$, $SE = 3.49$) than to complex non-speech analogues ($M = 63.5$, $SE = 4.31$) in the second experimental block ($t(21) = 2.84$, $p = .010$) (Figure 2). Means in the first block were not significantly different from each other ($t(21) = 1.18$, *ns*). The emergence of the effect of interest within the second block is consistent with other HAS studies.

Discussion

This study provides the first demonstration that human neonates are biased to listen to speech. This bias is

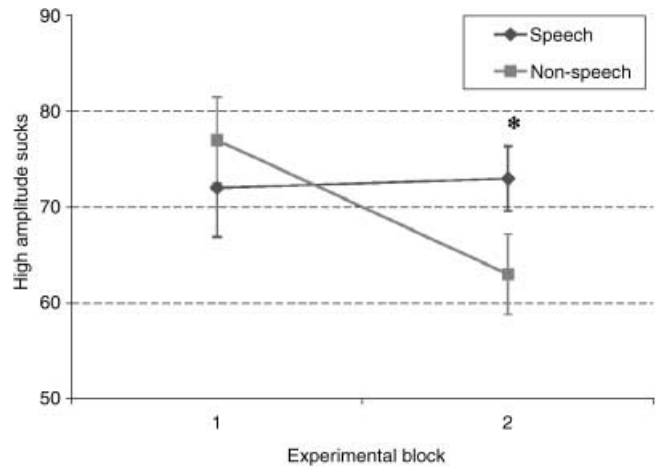


Figure 2 Neonates' HAS for speech and complex non-speech sounds. Neonates sucked significantly more to listen to speech than to complex non-speech analogues in the second experimental block.

consistent with initial proclivities, widespread in the animal kingdom, that direct animals towards particular types of auditory and visual information (e.g. Gould & Marler, 1987; Johnson, Bolhuis & Horn, 1992; Lorenz, 1965; Marler, 1990; Ryan, Phelps & Rand, 2001). Previous research had shown that infants process speech differently than non-speech; for example, they are better able to discriminate languages when speech is played forwards rather than backwards (Mehler *et al.*, 1988; Ramus *et al.*, 2000) even when the speech is low pass filtered (Mehler *et al.*, 1988; Nazzi *et al.*, 1998), or when every syllable is replaced with the consonant-vowel sequence /sa/ (Ramus, 2002), and they recruit differential neural resources for speech and non-speech processing (Dehaene-Lambertz *et al.*, 2002; Peña *et al.*, 2003). The present research focuses on a different aspect of speech and non-speech processing: whether infants show a behavioural bias for listening to speech. We compare neonates' listening preferences for speech to complex non-speech stimuli, and find a bias for listening to speech. This bias may confer an adaptive advantage by tuning humans to the communication signal of their conspecifics, and hence facilitate more in-depth processing and rapid learning of the specific attributes of the native language.

More than simply conferring an advantage, there is some evidence that a bias for speech in infancy may be essential for developing normal language abilities, as recent studies demonstrate that children with autism spectrum disorder (ASD) fail to show a preference for speech when it is compared with either an unresolvable stimulus composed of superimposed voices (Klin, 1991) or a complex non-speech stimulus composed of three sine-waves, similar to the non-speech counterpart of the current study (Klin, 1991; Kuhl, Coffey-Corina, Padden & Dawson, 2005). Instead, children

with ASD seem to show a preference for non-speech, the degree of which correlates significantly with ASD symptomatology, especially with respect to expressive language abilities (Kuhl *et al.*, 2005). Although this evidence is necessarily correlational, it is suggestive of the important role a bias for speech may play in normal language development.

The discovery of a neonatal bias for speech suggests several important questions. First, whence originates this neonatal bias for speech: is the bias rooted in prenatal experience with speech, or is it experience-independent? Previous studies have revealed neonatal preferences that are clearly experience-based, such as an attraction to the mother's voice (DeCasper & Fifer, 1980) and to the native language (Mehler, Bertoncini & Barriere, 1978; Moon, Cooper & Fifer, 1993). Though these specific preferences are unequivocally experience-based, a more general bias for speech, in its potential human universality, may not be. Indeed, evidence for innate conspecific preferences in other species suggests that a bias for speech might be a tantalizing candidate for an experience-independent human bias.

Second, what aspect of speech is the bias based on? In the case of duckling preference for conspecific calls, specific spectral and temporal aspects of the duck call are crucial. When conspecific calls are contrasted with a heterospecific foil similar in repetition rate and fundamental frequency, ducklings show no preference (Gottlieb, 1997). The neonatal speech preference could be based on a number of dimensions, ranging from low-level acoustic properties to higher-level abstract properties. For example, the sheer complexity of an acoustic stimulus can drive newborn and foetal physiological arousal for sounds; stimuli rich in spectral characteristics or patterned in temporal properties elicit greater changes in EMG (Hutt, Hutt, Lenard, van Bernuth & Muntjewerff, 1968), EEG (Lenard, von Bernuth & Hutt, 1969), and heart rate (Clarkson & Berg, 1983; Groome, Mooney, Holland, Smith, Atterbury & Dykman, 2000). Like other biologically special stimuli, such as faces or biological motion, attempts to create non-biological analogues must by necessity eliminate at least some of the characteristics of the original stimulus. Questions always remain as to whether the particular characteristics that were eliminated were the most important to maintain, or whether in mimicking the biological signal, one particular cue was highlighted over others. Though our complex non-speech stimuli preserve many of the spectral and temporal aspects of speech, they were composed of narrow frequency bands and thus necessarily lacked the broadband frequency information of speech. The relative acoustic complexity of speech may thus contribute to the neonatal preference observed in the current study. However, a preference for speech might stem from higher-level aspects of the speech stimulus, such as its human source, its bio-

logical origin or its intention to communicate. Studies are under way to investigate these possibilities.

Despite these remaining questions, a neonatal bias for speech is one important tool available at birth for learning language. Initial biases may be elaborated by experience to refine the perceptual preferences of developing organisms (e.g. Gottlieb, 1997; Werker & Tees, 1992). Indeed, the attraction to speech persists into the first few months of life (Vouloumanos & Werker, 2004) and may include communicative gestures in other modalities such as signed language (Krentz & Corina, *in press*). A speech bias, combined with established experience-based preferences for the mother's voice and native language, could provide human neonates with powerful tools for selecting and learning about communication signals from their rich environment.

Acknowledgements

This research was funded by fellowships from the Natural Science and Engineering Council (NSERC) of Canada, the Killam Foundation, and the Canadian Institutes of Health Research to Athena Vouloumanos, and grants from NSERC and the Human Frontiers Science Program, as well as a Canada Research Chair to Janet F. Werker. We thank M. Bhatnagar for her seminal role in implementing the project and her help with testing infants. Thanks to S. Bird and G. Carden for synthesis of non-speech stimuli, to T. Burns, E. Moon, and A. Valji for their help in testing infants, to the staff of the BC Women's and Children's Hospital for their cooperation, and to C. Narayan for assistance with the figure. We thank G. Marcus for invaluable discussions and comments on earlier versions. We especially thank all the parents and newborns who participated in our research.

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